The value of energy efficiency in Tunisia's power system (2030)

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Model: PyPSA-Earth v0.10.0 with the study code in this repository. All result numbers come from results/summary.csv, results/avoided.csv, results/pathway_*.csv, results/saving_cost.csv and results/validation.csv; all inputs come from data/assumptions_register.xlsx.

Summary

Tunisia plans 35% renewable electricity by 2030 and generates almost all the rest from gas (94.5% of generation in 2024). This study asks what each MWh of electricity saved, or moved in time, is worth to the 2030 power system.

Saving 15% in every hour (S1-15 vs S0) avoids, per MWh saved	Value
System cost	60 EUR
Gas burnt for power	0.91 MWh (thermal)
CO2	0.18 t
New capacity	449 MW per TWh saved per year

- With STEG's 10% reserve margin, the base case needs 423 MW of new gas turbines. A 10% saving already avoids all of them.
- Saving first on summer air conditioning (S1-15C) avoids 20% more gas per MWh than an even cut, although only 39% of the saving fits on the cooling load.
- Letting 300 MW of load move by up to 3 hours, on top of the 15% saving, cuts another 22 M EUR per year and 0.24 Mt of CO2.
- At the base gas price the least-cost system reaches 56% renewables, so the 35% target does not bind. It binds only in G-lo-S0, G-lo-S1-15.
- Built year by year at Tunisia's demonstrated pace (2027-2030 pathway), renewables reach only 22% in 2030 (27% with the 15% saving), and each MWh saved avoids 0.39 t of CO2 and 70 EUR.
- Saving is not free: a more efficient air conditioner costs 50-92 EUR per MWh saved, against 86 EUR avoided at the meter. Raising the standard to the EU's 2014 level pays for the power system.

1. Question and scope

The study measures what one MWh of electricity saved through efficiency, or moved in time through demand flexibility, avoids in 2030: new capacity, gas burnt, system cost and CO2. The renewable target stays in force in every scenario, so the comparison is between systems that all meet it.

The model covers Tunisia only, with two network nodes (north and south), 8,760 hourly steps and the 2013 weather year, solved with the open-source solver HiGHS. It does not include the cost of the efficiency programmes themselves, the planned ELMED link to Italy, or other countries.

2. Method

2.1 Base network and 2024 calibration

PyPSA-Earth, used unmodified, builds the Tunisian grid, power plants, renewable potentials and hourly demand from open data. Its plant list does not match the official fleet, so the thermal fleet is rebuilt from the ONEM 2024 statistics: 2,527 MW of combined cycle and 3,117 MW of gas turbines and steam units. One efficiency factor for existing gas units is fitted so that a 2024 run burns as much gas as STEG reported. The hourly demand shape is stretched to the official load factor and minimum-to-peak ratio.

2024 check (calibrated model)	Model	Official	Difference
Gas burnt for power (TWh thermal)	37.0	36.8	+0.4%
Gas fleet efficiency (net)	0.49	0.49	+0.0%
Peak demand (MW)	4,901	4,888	+0.3%
Gas share of generation	93.6%	94.5%	-1.0%

The gas-burnt check is independent of the fit: the factor was fitted on STEG's specific consumption, and the gas total comes from a separate ONEM table.

2.2 The 2030 system

Demand grows from 22.5 TWh in 2024 at 3.2% per year, STEG's base-case forecast as quoted by JICA (2022). Imports stay at their 2024 level (3.1 TWh, flat over the year), so every MWh saved falls on domestic generation. The model then builds solar, onshore wind, batteries and gas turbines at least cost, using 2030 costs from PyPSA technology-data. In every scenario it must:

- produce at least 35% of domestic generation from solar, wind and hydro;
- keep firm capacity (gas and hydro) at least 10% above peak demand, STEG's planning margin;
- build the committed solar and wind projects (data/committed_projects.csv) and keep the existing fleet.

After each solve the pipeline checks the energy balance, the renewable share, the reserve margin, that no load is shed and that committed and existing capacity is kept. All 11 scenarios pass.

2.3 Scenarios

Scenario	What changes against S0
S0	Base case: 2030 demand, base gas price
S1-10, S1-15, S1-20	Demand 10, 15 or 20% lower in every hour
S1-15C	Same 15% of annual energy, taken first from summer cooling load
S2-300, S2-600	S1-15 plus 300 or 600 MW of load that can move by up to 3 hours
G-lo-S0, G-lo-S1-15	S0 and S1-15 with the gas price x0.5
G-hi-S0, G-hi-S1-15	S0 and S1-15 with the gas price x1.5

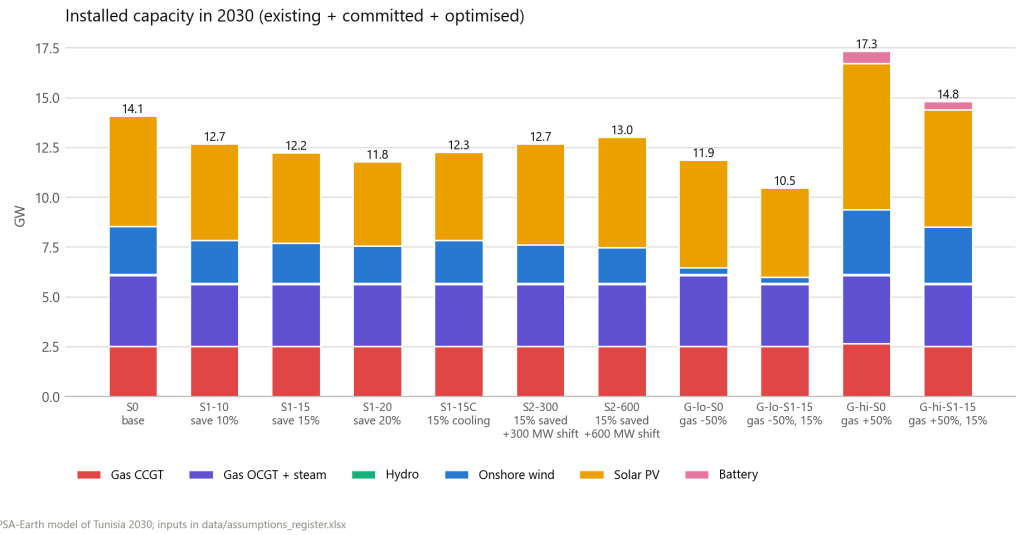
Shiftable load is modelled as a free, lossless store at each node that starts and ends every day half full. Load can be brought forward or postponed, but every MWh moved must come back within 3 hours, and the load moved up plus the load moved down in one hour cannot exceed the shift limit.

3. Results

3.1 Main scenarios

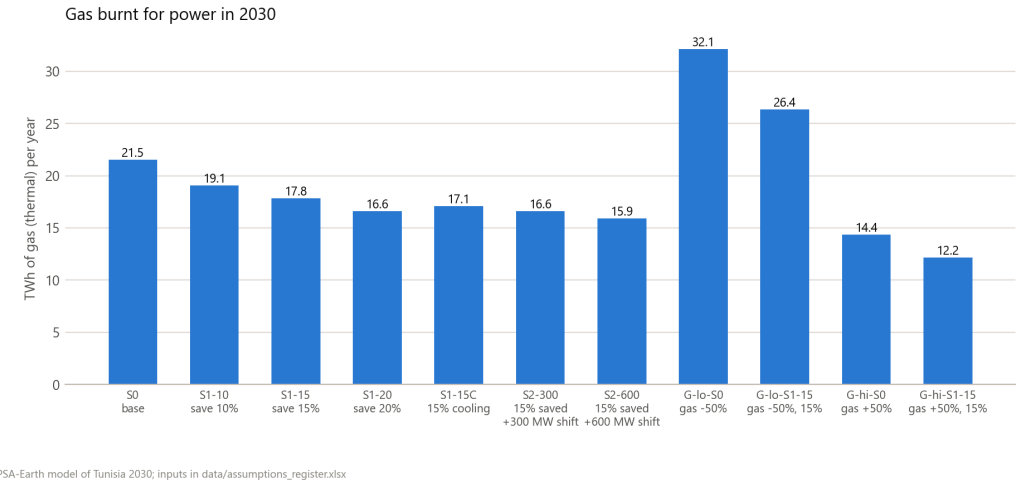
	S0	S1-10	S1-15	S1-20	S1-15C	S2-300	S2-600
Domestic demand (TWh)	24.08	21.36	20.00	18.64	20.00	20.00	20.00

	S0	S1-10	S1-15	S1-20	S1-15C	S2-300	S2-600
New gas turbines (MW)	423	0	0	0	0	0	0
New solar (MW)	3,797	3,105	2,796	2,474	2,670	3,321	3,816
New onshore wind (MW)	2,121	1,857	1,714	1,574	1,853	1,629	1,485
Gas burnt (TWh thermal)	21.55	19.09	17.83	16.60	17.09	16.63	15.89
CO2 (Mt)	4.27	3.78	3.53	3.29	3.38	3.29	3.15
Renewable share	56%	56%	56%	56%	57%	58%	60%
System cost (M EUR/yr)	1,327	1,157	1,084	1,012	1,073	1,062	1,045
Average price (EUR/MWh)	54.7	54.2	53.5	52.8	53.1	52.5	51.9



Source: own PyPSA-Earth model of Tunisia 2030; inputs in data/assumptions_register.xlsx

Figure 1. Installed capacity in 2030 by scenario.



Source: own PyPSA-Earth model of Tunisia 2030; inputs in data/assumptions_register.xlsx

Figure 2. Gas burnt for power in 2030 by scenario.

3.2 What one MWh saved avoids

Scenario	vs	Cost (EUR/MWh)	CO2 (t/MWh)	Gas (MWh/MWh)	Capacity (MW per TWh)
S1-10	S0	62.6	0.18	0.90	507

Scenario	vs	Cost (EUR/MWh)	CO2 (t/MWh)	Gas (MWh/MWh)	Capacity (MW per TWh)
S1-15	S0	59.6	0.18	0.91	449
S1-20	S0	58.0	0.18	0.91	422
S1-15C	S0	62.4	0.22	1.09	446
S2-300	S0	65.0	0.24	1.21	341
S2-600	S0	69.2	0.27	1.39	255
G-lo-S1-15	G-lo-S0	43.0	0.28	1.42	348
G-hi-S1-15	G-hi-S0	70.1	0.11	0.54	616

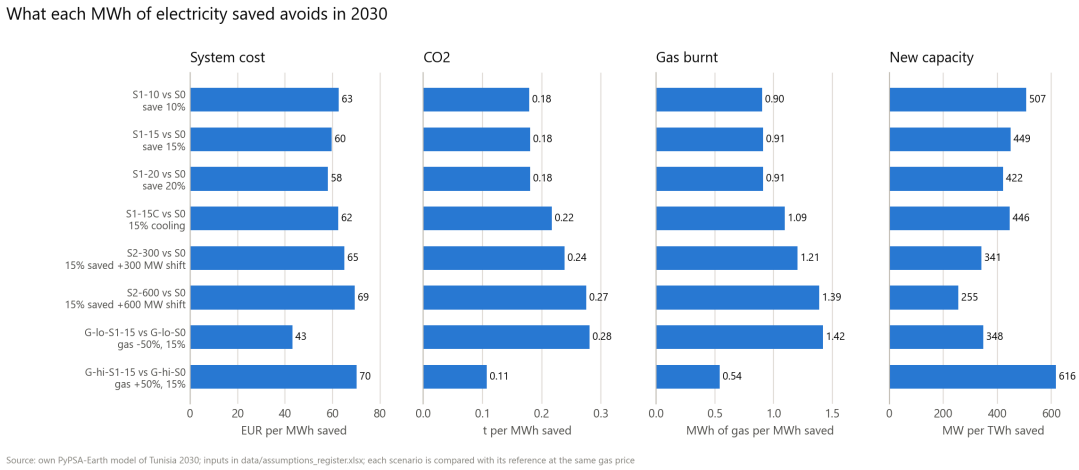


Figure 3. What each MWh saved avoids, against the reference at the same gas price.

3.3 Findings

Efficiency avoids new gas plant. The base case needs 423 MW of new gas turbines to keep the reserve margin. A 10% saving lowers the peak enough to avoid all of it, which is why the first 10% saved is worth the most (62.6 EUR/MWh against 59.6 for 15%). Beyond that point, further savings avoid mostly solar and wind that would otherwise be built.

Timing matters. In S1-15C only 39% of the 15% saving fits on the summer cooling load; the rest is an even cut. That is enough to avoid 20% more gas per MWh and 11.5 M EUR per year more than S1-15, because air-conditioning hours are the hours when gas runs.

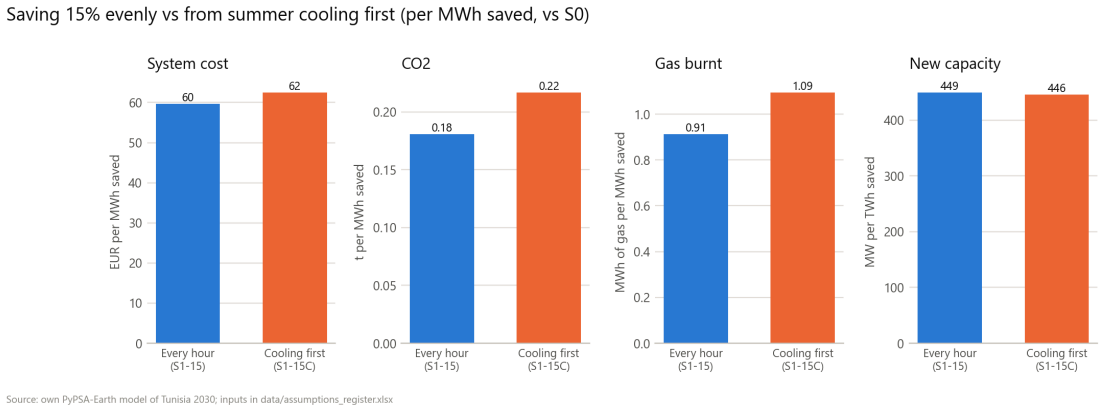


Figure 4. The same 15% saved evenly or from cooling first.

Flexibility adds to efficiency. Shifting 300 MW of load saves 22.1 M EUR per year on top of S1-15, about 74,000 EUR per MW per year. The model moves morning and evening load into the midday solar hours, builds more solar and burns less gas. Under the reserve rule, flexibility gets no credit for lowering the peak, so all of this value is energy.

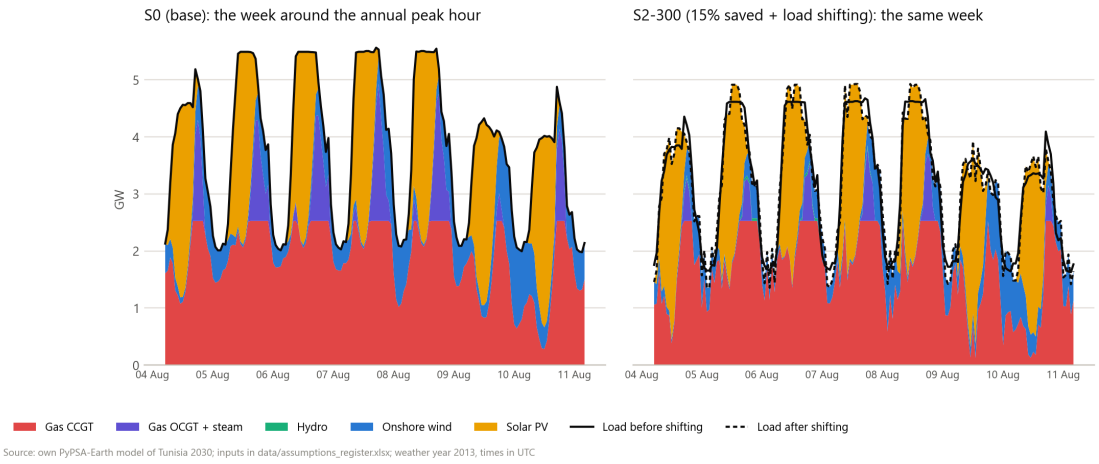


Figure 5. Dispatch in the week around the annual peak, S0 and S2-300.

The gas price changes the result most. With gas at x0.5, each MWh saved avoids 0.28 t of CO2, because the saving mostly displaces gas. With gas at x1.5, the reference system is already 70% renewable with 3,148 MWh of batteries, and each MWh saved avoids only 0.11 t of CO2 but 70 EUR.

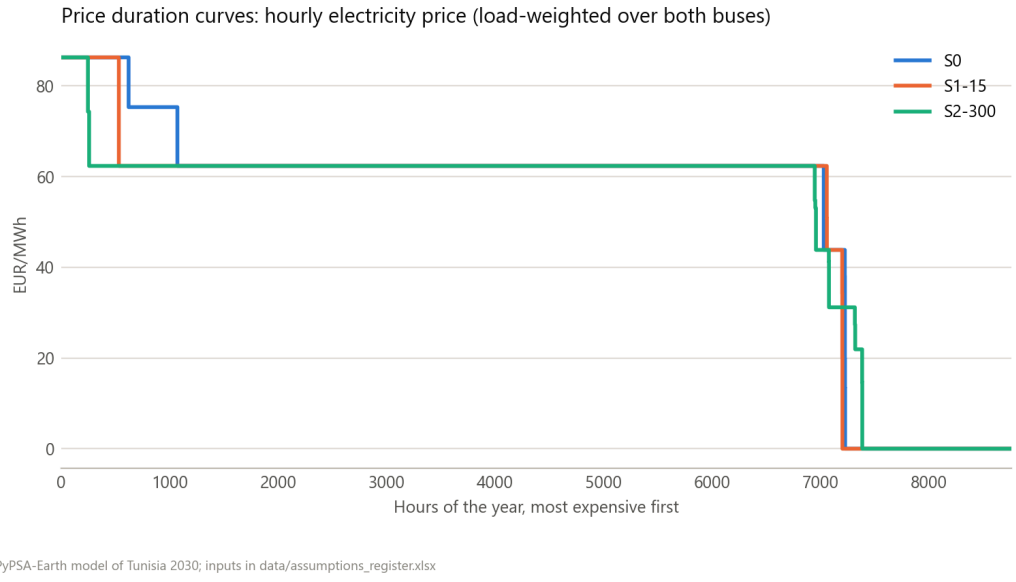


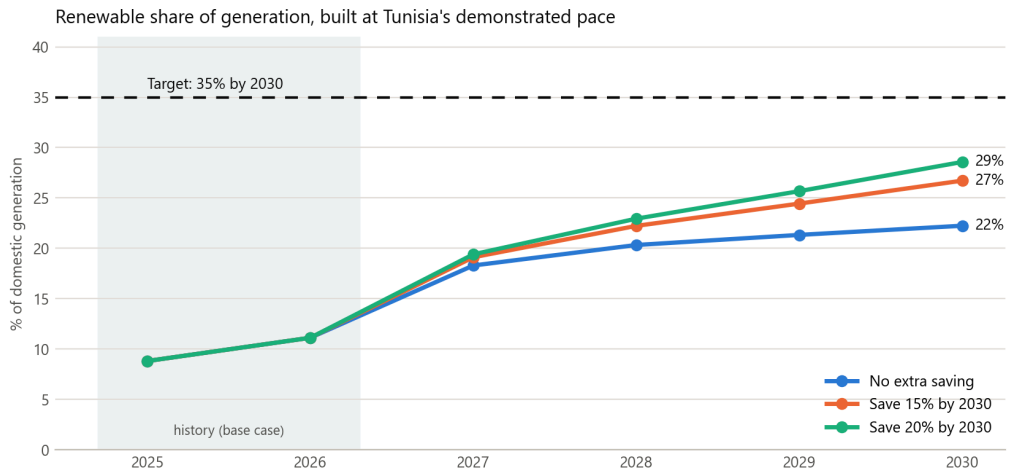
Figure 6. Hourly electricity price, sorted, for S0, S1-15 and S2-300.

3.4 The 2027-2030 pathway

The runs above solve 2030 in one step and may build any amount. The pathway instead solves 2025 and 2026 once (base case), then each scenario for 2027, 2028, 2029 and 2030 in turn. Each year keeps what was built before, committed projects switch on in their expected year, and new building is capped at Tunisia's demonstrated pace: 268 MW of solar a year beyond the committed projects, no new wind before 2029 and then 75 MW a year, and batteries only from 2029 (at most 150 MW / 540 MWh). Savings and shifting reach their full size in 2030, a quarter at a time from 2027. The 35% target is measured, not enforced.

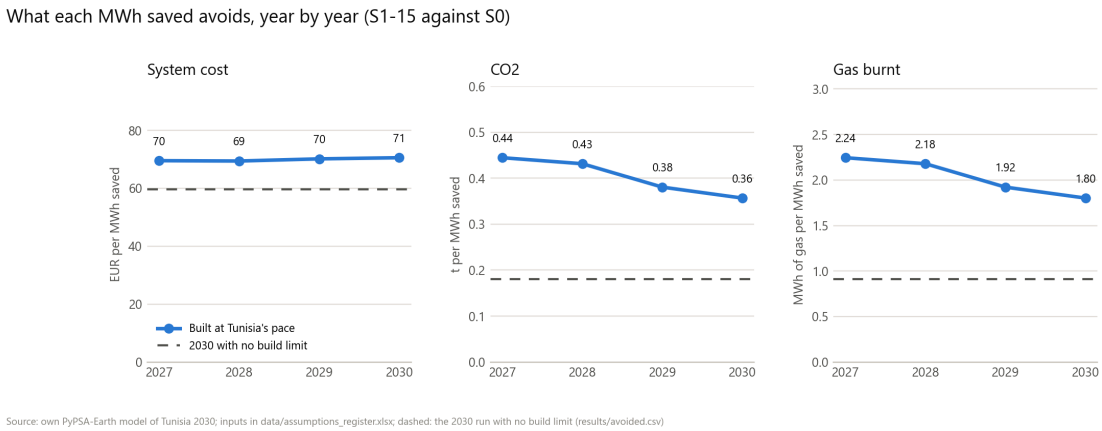
Over 2027-2030, per MWh saved	Cost (EUR)	CO2 (t)	Gas (MWh)
S1-10 vs S0	71.1	0.36	1.83
S1-15 vs S0	70.1	0.39	1.95
S1-20 vs S0	69.3	0.40	2.00
S2-300 vs S0	72.0	0.40	2.02
G-lo-S1-15 vs G-lo-S0	40.9	0.43	2.19
G-hi-S1-15 vs G-hi-S0	95.2	0.36	1.80

- Renewables reach 22% of generation in 2030 without extra saving, 27% when saving 15% and 29% when saving 20%: the 35% target is out of reach at this pace, and efficiency closes part of the gap.
- Each MWh saved (S1-15) avoids 0.39 t of CO2 and 1.95 MWh of gas, against 0.18 t and 0.91 MWh in the unconstrained 2030 run: with little room to build solar, a saved MWh comes off a gas plant.
- Without extra saving the model adds 423 MW of combined cycles by 2030 for the reserve margin; every saving scenario adds none.
- Shifting 300 MW is worth 6.8 M EUR in 2030, much less than in the unconstrained run, because there is little midday solar to shift into.



Source: own PyPSA-Earth model of Tunisia 2030; inputs in data/assumptions_register.xlsx; build limits in the register (pathway_*)

Figure 7. Renewable share year by year on the pathway, against the target.



Source: own PyPSA-Earth model of Tunisia 2030; inputs in data/assumptions_register.xlsx; dashed: the 2030 run with no build limit (results/avoided.csv)

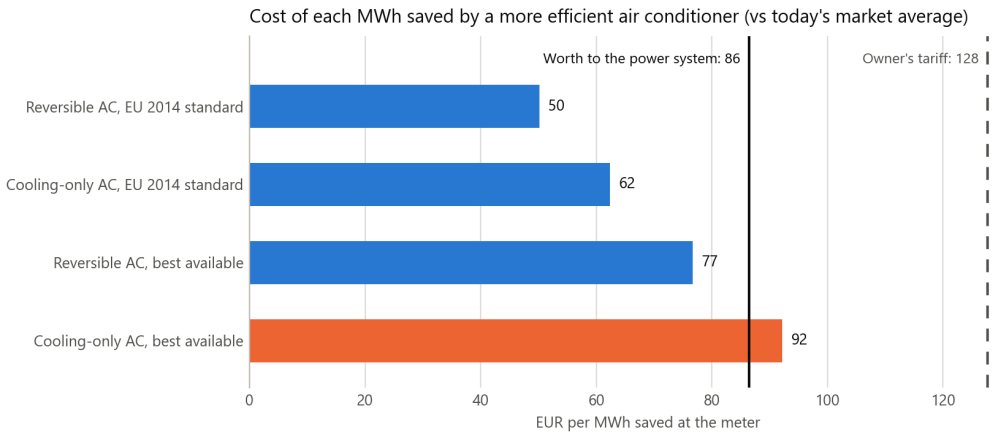
Figure 8. What each MWh saved avoids, year by year (S1-15 vs S0), against the unconstrained 2030 run (dashed).

3.5 What saving costs: efficient air conditioners

The model treats saving as free. The one Tunisian measure with a published cost per kWh saved is the split air conditioner (LBNL, Cost-Benefit Analysis for Air Conditioner in Tunisia, 2023): prices and yearly electricity use at each efficiency level, against today's market average. The extra price is spread over the unit's 10.5-year life at the model's 7.1% discount rate and divided by the kWh saved each year, then converted at 1.1827 US$ per EUR (2021). It is compared with the value of a cooling-first MWh saved on the pathway (S1-15C, 2027-2030), raised by the 18% grid losses because the AC saves at the meter, and with the STEG tariff the owner stops paying.

Measure (vs today's market average)	Extra price (US$)	kWh saved a year	Cost (EUR/MWh)	Net for the system (EUR/MWh)
Cooling-only AC, EU 2014 standard	48	90	62	+24
Cooling-only AC, best available	220	279	92	-6
Reversible AC, EU 2014 standard	51	119	50	+36
Reversible AC, best available	227	346	77	+10

- A saved MWh is worth 86 EUR to the power system at the meter (71 EUR at the plants) and 128 EUR to the owner.
- Raising the minimum standard to the EU's 2014 level costs 50-62 EUR per MWh saved: it pays for the power system.
- The best available units cost 77-92 EUR per MWh: close to break-even for the system, and every level pays for the owner.
- A standard would deliver 9% (EU level) to 25% (best available) of the S1-15C saving in 2030; the other measures have no sourced Tunisian cost per kWh.



Source: LBNL 2023 (prices and kWh per unit); value: own model, S1-15C 2027-2030 plus 18% grid losses; results/saving_cost.csv

Figure 9. Cost per MWh saved by more efficient air conditioners, against what the MWh is worth to the power system and to the owner.

4. Limitations

- In the 2030 runs nothing limits how fast renewables are built: the base case adds 5.9 GW of solar and wind, far above Tunisia's recent pace. The pathway (section 3.4) caps building; its caps are judgements based on past pace, and it assumes committed projects arrive on time.

- The cost of efficiency programmes is not modelled. The avoided cost per MWh is the most a programme could cost and still pay off for the power system. Section 3.5 prices one measure (air-conditioner standards, equipment cost only); the cost of the whole saving is unknown.
- One weather year (2013) at 1-hour steps. A 3-hour test run built 2.7% more solar and 3.9% more wind (resolution benchmark in docs/project-history/phase2-base-network.md).
- Demand growth and the 2030 peak come from STEG's forecast as quoted by JICA (2022), a secondary source.
- Imports are flat at their 2024 level, so every MWh saved falls on domestic generation.
- The fitted efficiency factor for existing gas units also stands in for plant outages, which are not modelled.
- The reserve margin is one national capacity rule, not a reliability study.
- System cost includes the annualised cost of plants that already exist. Differences between scenarios are meaningful; the absolute level is not the cost of the real system.

5. Inputs and sources

Every input is a row of data/assumptions_register.xlsx with its unit, type and source; rows typed 'assumption' have no published source and are explained in the register notes. The study inputs:

Input	Value	Unit	Type	Source
re_target_2030	0.35	share	sourced	TAP, 9 Oct 2026: Head of Government reaffirms 35% target
flex_cap_mw	300	MW	sourced	TAP, 9 Oct 2026: action plan targets 300 MW peak-demand reduction
flex_cap_hi_mw	600	MW	assumption	
flex_max_hours	3	h	assumption	
gas_price_factor_lo	0.5	x base	assumption	
gas_price_factor_hi	1.5	x base	assumption	
demand_growth	0.032	1/yr	sourced	JICA / TEPSCO / KPMG, Data Collection Survey on Power Sector in Tunisia, Final Report (Mar 2022), quoting STEG's base-case forecast, p.2-17 and p.5-34
peak_ref_2030_mw	6,000	MW	sourced	JICA / TEPSCO / KPMG, Data Collection Survey on Power Sector in Tunisia, Final Report (Mar 2022), quoting STEG's base-case forecast, p.2-17 and p.5-35
discount_rate	0.071	1/yr	sourced	PyPSA-Earth v0.10.0 config.default.yaml, costs.discountrate
demand_2024_twh	22.51	TWh	sourced	ONEM / Ministry of Industry, Mines and Energy, Memento du secteur de l'énergie, Chiffres de 2024 (Aug 2025), table 'ELECTRICITE', printed p.18
imports_2024_twh	3.113	TWh	sourced	ONEM / Ministry of Industry, Mines and Energy, Memento du secteur de l'énergie, Chiffres de 2024 (Aug 2025), table 'ELECTRICITE', printed p.18

Input	Value	Unit	Type	Source
gas_price_eur_per_mwh_th	28.42	EUR/MWh_th	sourced	PyPSA technology-data v0.15.0 costs_2030.csv (gas, fuel); original: World Bank Commodity Markets 2025 and ENTSO-E/ENTSOG TYNDP 2024
cap_2024_ccgt_mw	2,527	MW	sourced	ONEM / Ministry of Industry, Mines and Energy, Memento du secteur de l'énergie, Chiffres de 2024 (Aug 2025)
cap_2024_ocgt_mw	2,457	MW	sourced	ONEM / Ministry of Industry, Mines and Energy, Memento du secteur de l'énergie, Chiffres de 2024 (Aug 2025)
cap_2024_steam_mw	660	MW	sourced	ONEM / Ministry of Industry, Mines and Energy, Memento du secteur de l'énergie, Chiffres de 2024 (Aug 2025)
cap_2024_re_mw	368	MW	sourced	ONEM / Ministry of Industry, Mines and Energy, Memento du secteur de l'énergie, Chiffres de 2024 (Aug 2025)
cap_2024_total_mw	6,012	MW	sourced	ONEM / Ministry of Industry, Mines and Energy, Memento du secteur de l'énergie, Chiffres de 2024 (Aug 2025)
gen_2024_gas_twh	18.16	TWh	sourced	ONEM / Ministry of Industry, Mines and Energy, Memento du secteur de l'énergie, Chiffres de 2024 (Aug 2025)
gen_2024_oil_twh	0.0719	TWh	sourced	ONEM / Ministry of Industry, Mines and Energy, Memento du secteur de l'énergie, Chiffres de 2024 (Aug 2025)
gen_2024_hydro_twh	0.016	TWh	sourced	ONEM / Ministry of Industry, Mines and Energy, Memento du secteur de l'énergie, Chiffres de 2024 (Aug 2025)
gen_2024_wind_twh	0.337	TWh	sourced	ONEM / Ministry of Industry, Mines and Energy, Memento du secteur de l'énergie, Chiffres de 2024 (Aug 2025)
gen_2024_solar_twh	0.6278	TWh	sourced	ONEM / Ministry of Industry, Mines and Energy, Memento du secteur de l'énergie, Chiffres de 2024 (Aug 2025)
gen_2024_total_twh	19.21	TWh	sourced	ONEM / Ministry of Industry, Mines and Energy, Memento du secteur de l'énergie, Chiffres de 2024 (Aug 2025)
gas_share_2024	0.945	share	sourced	ONEM / Ministry of Industry, Mines and Energy, Memento du secteur de l'énergie, Chiffres de 2024 (Aug 2025)
peak_2024_mw	4,888	MW	sourced	ONEM / Ministry of Industry, Mines and Energy, Memento du secteur de l'énergie, Chiffres de 2024 (Aug 2025)
steg_specific_consumption_2024	194.8	toe(GCV)/GWh	sourced	ONEM / Ministry of Industry, Mines and Energy, Memento du secteur de l'énergie, Chiffres de 2024 (Aug 2025)

Input	Value	Unit	Type	Source
steg_demand_2030_gwh	28,362	GWh	sourced	JICA / TEPSCO / KPMG, Data Collection Survey on Power Sector in Tunisia, Final Report (Mar 2022), quoting STEG's base-case forecast, p.2-17
load_factor_2030	0.54	share	sourced	JICA / TEPSCO / KPMG, Data Collection Survey on Power Sector in Tunisia, Final Report (Mar 2022), quoting STEG's base-case forecast, p.5-34
min_to_peak_2030	0.34	share	sourced	JICA / TEPSCO / KPMG, Data Collection Survey on Power Sector in Tunisia, Final Report (Mar 2022), quoting STEG's base-case forecast, p.5-35
ccgt_efficiency	0.58	per unit	sourced	PyPSA technology-data v0.15.0 costs_2030.csv (CCGT, efficiency); original: Danish Energy Agency, inputs/technology_data_for_el_and_dh.xlsx
ccgt_vom	5.61	EUR/MWh	sourced	PyPSA technology-data v0.15.0 costs_2030.csv (CCGT, VOM); original: Danish Energy Agency, inputs/technology_data_for_el_and_dh.xlsx
ocgt_efficiency	0.41	per unit	sourced	PyPSA technology-data v0.15.0 costs_2030.csv (OCGT, efficiency); original: Danish Energy Agency, inputs/technology_data_for_el_and_dh.xlsx
ocgt_vom	6.011	EUR/MWh	sourced	PyPSA technology-data v0.15.0 costs_2030.csv (OCGT, VOM); original: Danish Energy Agency, inputs/technology_data_for_el_and_dh.xlsx
gas_ncv_per_gcv	0.9	ratio	sourced	ONEM / Ministry of Industry, Mines and Energy, Memento du secteur de l'énergie, Chiffres de 2024, conversion table (PDF p.22)
existing_gas_efficiency_factor	0.8634	x tech-data	assumption	Calibrated (Task 10) so the dispatch-only 2024 run matches STEG's measured fuel use: ONEM Memento 2024, specific consumption 194.8 toe(PCS)/GWh, PCI = 0.9 PCS
ocgt_investment	581.4	EUR/kW	sourced	PyPSA technology-data v0.15.0 costs_2030.csv (OCGT, investment); original: Danish Energy Agency, inputs/technology_data_for_el_and_dh.xlsx
ocgt_fom	1.78	%/year	sourced	PyPSA technology-data v0.15.0 costs_2030.csv (OCGT, FOM); original: Danish Energy Agency, inputs/technology_data_for_el_and_dh.xlsx
ocgt_lifetime	25	years	sourced	PyPSA technology-data v0.15.0 costs_2030.csv (OCGT, lifetime); original: Danish Energy Agency, inputs/technology_data_for_el_and_dh.xlsx

Input	Value	Unit	Type	Source
gas_co2_intensity	0.198	tCO2/MWh_th	sourced	PyPSA technology-data v0.15.0 costs_2030.csv (gas, CO2 intensity); original: Stoichiometric calculation with 50 GJ/t CH4
gas_for_power_2024_ktep_ncv	3,168	ktep (net CV)	sourced	ONEM / Ministry of Industry, Mines and Energy, Memento du secteur de l'énergie, Chiffres de 2024 (Aug 2025)
cap_2024_selfgen_solar_lv_mw	315	MW	sourced	ONEM / Ministry of Industry, Mines and Energy, Memento du secteur de l'énergie, Chiffres de 2024 (Aug 2025)
cap_2024_selfgen_solar_mvhv_mw	125	MW	sourced	ONEM / Ministry of Industry, Mines and Energy, Memento du secteur de l'énergie, Chiffres de 2024 (Aug 2025)
reserve_margin	0.1	share of peak	sourced	JICA / TEPSCO / KPMG, Data Collection Survey on Power Sector in Tunisia, Final Report (Mar 2022), p.5-33
gas_supply_domestic_2024_ktep	1,212	ktep (net CV)	sourced	ONEM / Ministry of Industry, Mines and Energy, Memento du secteur de l'énergie, Chiffres de 2024 (Aug 2025), table 'Gaz naturel' (approvisionnement, ktep-pci)
gas_supply_royalty_2024_ktep	909	ktep (net CV)	sourced	ONEM / Ministry of Industry, Mines and Energy, Memento du secteur de l'énergie, Chiffres de 2024 (Aug 2025), table 'Gaz naturel' (approvisionnement, ktep-pci)
gas_supply_purchases_2024_ktep	2,290	ktep (net CV)	sourced	ONEM / Ministry of Industry, Mines and Energy, Memento du secteur de l'énergie, Chiffres de 2024 (Aug 2025), table 'Gaz naturel' (approvisionnement, ktep-pci)
energy_independence_2024	0.41	share	sourced	ONEM / Ministry of Industry, Mines and Energy, Memento du secteur de l'énergie, Chiffres de 2024 (Aug 2025), 'Bilan d'énergie primaire'
steg_established_year	1,962	year	sourced	JICA / TEPSCO / KPMG, Data Collection Survey on Power Sector in Tunisia, Final Report (Mar 2022), p.2-1
steg_generation_share	0.86	share	sourced	JICA / TEPSCO / KPMG, Data Collection Survey on Power Sector in Tunisia, Final Report (Mar 2022), p.2-1
anme_established_year	1,985	year	sourced	JICA / TEPSCO / KPMG, Data Collection Survey on Power Sector in Tunisia, Final Report (Mar 2022), p.2-1
tsp_re_target_2030	0.3	share	sourced	JICA / TEPSCO / KPMG, Data Collection Survey on Power Sector in Tunisia, Final Report (Mar 2022), p.2-2
indc_co2_intensity_cut_2030	0.41	share	sourced	JICA / TEPSCO / KPMG, Data Collection Survey on Power Sector in Tunisia, Final Report (Mar 2022), p.2-2

Input	Value	Unit	Type	Source
gas_supply_domestic_2023_ktep	1,607	ktep (net CV)	sourced	ONEM / Ministry of Industry, Mines and Energy, Memento du secteur de l'énergie, Chiffres de 2024 (Aug 2025), table 'Gaz naturel' (approvisionnement, ktep-pci)
solar_capacity_2024_mw	773	MW	sourced	Enerdata, 'France's Voltalia expands Tunisian portfolio with 132 MW solar project' (4 Feb 2026)
solar_growth_2024	0.53	share	sourced	Enerdata, 'France's Voltalia expands Tunisian portfolio with 132 MW solar project' (4 Feb 2026)
pathway_solar_extra_mw_per_year	268	MW/yr	assumption	
pathway_wind_first_year	2,029	year	assumption	
pathway_wind_extra_mw_per_year	75	MW/yr	assumption	Enerdata, 'Tunisia launches a tender for 1.7 GW of renewables' (5 Jan 2023)
pathway_battery_first_year	2,029	year	assumption	Mercom, 'Tunisia invites bids for 300 MW solar plus 150 MW/540 MWh battery storage project' (Mar 2026)
pathway_battery_mw	150	MW	assumption	Mercom, 'Tunisia invites bids for 300 MW solar plus 150 MW/540 MWh battery storage project' (Mar 2026)
anme_demand_cut_2030	0.3	share	sourced	AllAfrica / La Presse, ANME announcement of the national energy-conservation strategy (24 Oct 2025)
pathway_battery_mwh	540	MWh	assumption	Mercom, 'Tunisia invites bids for 300 MW solar plus 150 MW/540 MWh battery storage project' (Mar 2026)
ac_lifetime_years	10.5	years	sourced	LBNL (Letschert, Agarwal, de la Rue du Can, Park), Cost-Benefit Analysis for Air Conditioner in Tunisia, LBNL-2001527 (Feb 2023), Table 13
ac_cool_price_bau_usd	469	US$	sourced	LBNL (Letschert, Agarwal, de la Rue du Can, Park), Cost-Benefit Analysis for Air Conditioner in Tunisia, LBNL-2001527 (Feb 2023), Table 11
ac_cool_price_eu_usd	517	US$	sourced	LBNL (Letschert, Agarwal, de la Rue du Can, Park), Cost-Benefit Analysis for Air Conditioner in Tunisia, LBNL-2001527 (Feb 2023), Table 11
ac_cool_price_best_usd	689	US$	sourced	LBNL (Letschert, Agarwal, de la Rue du Can, Park), Cost-Benefit Analysis for Air Conditioner in Tunisia, LBNL-2001527 (Feb 2023), Table 11
ac_cool_kwh_bau	604	kWh/yr	sourced	LBNL (Letschert, Agarwal, de la Rue du Can, Park), Cost-Benefit Analysis for Air Conditioner in Tunisia, LBNL-2001527 (Feb 2023), Table 11

Input	Value	Unit	Type	Source
ac_cool_kwh_eu	514	kWh/yr	sourced	LBNL (Letschert, Agarwal, de la Rue du Can, Park), Cost-Benefit Analysis for Air Conditioner in Tunisia, LBNL-2001527 (Feb 2023), Table 11
ac_cool_kwh_best	325	kWh/yr	sourced	LBNL (Letschert, Agarwal, de la Rue du Can, Park), Cost-Benefit Analysis for Air Conditioner in Tunisia, LBNL-2001527 (Feb 2023), Table 11
ac_rev_price_bau_usd	486	US$	sourced	LBNL (Letschert, Agarwal, de la Rue du Can, Park), Cost-Benefit Analysis for Air Conditioner in Tunisia, LBNL-2001527 (Feb 2023), Table 12
ac_rev_price_eu_usd	537	US$	sourced	LBNL (Letschert, Agarwal, de la Rue du Can, Park), Cost-Benefit Analysis for Air Conditioner in Tunisia, LBNL-2001527 (Feb 2023), Table 12
ac_rev_price_best_usd	713	US$	sourced	LBNL (Letschert, Agarwal, de la Rue du Can, Park), Cost-Benefit Analysis for Air Conditioner in Tunisia, LBNL-2001527 (Feb 2023), Table 12
ac_rev_kwh_bau	887	kWh/yr	sourced	LBNL (Letschert, Agarwal, de la Rue du Can, Park), Cost-Benefit Analysis for Air Conditioner in Tunisia, LBNL-2001527 (Feb 2023), Table 12
ac_rev_kwh_eu	768	kWh/yr	sourced	LBNL (Letschert, Agarwal, de la Rue du Can, Park), Cost-Benefit Analysis for Air Conditioner in Tunisia, LBNL-2001527 (Feb 2023), Table 12
ac_rev_kwh_best	541	kWh/yr	sourced	LBNL (Letschert, Agarwal, de la Rue du Can, Park), Cost-Benefit Analysis for Air Conditioner in Tunisia, LBNL-2001527 (Feb 2023), Table 12
ac_tariff_usd_per_kwh	0.151	US$/kWh	sourced	LBNL (Letschert, Agarwal, de la Rue du Can, Park), Cost-Benefit Analysis for Air Conditioner in Tunisia, LBNL-2001527 (Feb 2023), Table 13
grid_losses	0.18	share	sourced	LBNL (Letschert, Agarwal, de la Rue du Can, Park), Cost-Benefit Analysis for Air Conditioner in Tunisia, LBNL-2001527 (Feb 2023), Table 16
ac_saving_2030_eu_gwh	287	GWh/yr	sourced	LBNL (Letschert, Agarwal, de la Rue du Can, Park), Cost-Benefit Analysis for Air Conditioner in Tunisia, LBNL-2001527 (Feb 2023), Table 17
ac_saving_2030_best_gwh	841	GWh/yr	sourced	LBNL (Letschert, Agarwal, de la Rue du Can, Park), Cost-Benefit Analysis for Air Conditioner in Tunisia, LBNL-2001527 (Feb 2023), Table 17

Input	Value	Unit	Type	Source
usd_per_eur_2021	1.183	US$/EUR	sourced	Deutsche Bundesbank, ECB euro reference rate, annual average, series BBEX3.A.USD.EUR.BB.AC.A04

Modelling tools. The study is built on these open-source tools:

- T. Brown, J. Hörsch, D. Schlachtberger, "PyPSA: Python for Power System Analysis", Journal of Open Research Software 6(1), 4 (2018). doi:10.5334/jors.188
- M. Parzen et al., "PyPSA-Earth. A new global open energy system optimization model demonstrated in Africa", Applied Energy 341, 121096 (2023). doi:10.1016/j.apenergy.2023.121096

6. Reproducing the study

The code, tests, scenario files and step-by-step instructions are in the repository README. One command (snakemake -j 1) rebuilds, solves and checks all 11 scenarios, for 2030 and for the 2027-2030 pathway, in about two hours on a laptop, and scripts/make_figures.py and scripts/make_report.py redraw the figures and this report.